

NATIONAL TRANSPORTATION SAFETY BOARD

Office of Research and Engineering
Materials Laboratory Division
Washington, D.C. 20594



December 7, 2006

MATERIALS LABORATORY FACTUAL REPORT

Report No. 06-079

A. ACCIDENT

Place : Boston, Massachusetts
Date : July 10, 2006
Vehicle : Connector Tunnel to Ted Williams Tunnel
NTSB No. : HWY06MH024
Investigator : Mark Bagnard

B. COMPONENTS EXAMINED

Anchor displacement data

C. DETAILS OF THE EXAMINATION

Background / Discussion

During on-scene portions of the investigation (July 14 to July 19 and August 8 to August 10, 2006), each adhesive anchor remaining in the roof of the connector tunnel was examined to measure the distance that the anchor had moved downward (displaced) from its originally installed position.¹ At the time of the examination, all hardware had been disassembled from the anchors. The anchor displacement distance was primarily based on the position of the red plastic cap that was inserted into the anchor hole prior to installation of the anchor. It should be noted that it was sometimes difficult to exactly determine the initial position of the red plastic cap. Most of the data was recorded to the nearest 1/16 of an inch.

The data is presented in both tabular form for each individual anchor, and in two graphical forms. Appendix 1 contains the tabular data for each row in each section of the West-Bound (WB), East-Bound (EB), and High Occupancy Vehicle (HOV) tunnels.² This appendix also contains bar charts used to represent downward displacement. In these charts, all data for the group of anchors from a specific hanger support plate connection

¹ The data for a few of the anchors was taken from Federal Highway Administration data gathered before the anchor was removed as part of their load testing.

² The averaged data contains numbers calculated out up to six decimal places, but these values do not represent actual accuracy to this level.

(either a two-anchor or four-anchor fitting) were averaged to generate a single number to represent the amount that the fitting would have moved downward. Bar charts are not shown for some of the rows with minimal or no anchor displacement.

The second graphic form, see figure 1, shows an overall view of all of the data for each of the three tunnels in order to show general relationships. The displays in figure 1 use the same average anchor group displacement data contained in the tables in appendix 1, but with the shading of a square box representing the extent to which the fitting represented by the box was displaced downward. Darker shades represent greater downward displacement.

Data Summary

The following three tables summarize the number of displaced anchors in the anchor group rows and modules for the WB, EB, and HOV tunnels.

WB Tunnel	Number of Displaced Anchors/Total Anchors				
	Module 081	Module 082	Module 083	Module 084	Total
North Row	3/10	6/18	9/18	2/18	20/64
Middle Row	7/10	11/20	18/20	18/20	54/70
South Row	0/4	0/18	0/18	4/18	4/58
Total	10/24	17/56	27/56	24/56	78/192

EB Tunnel	Number of Displaced Anchors/Total Anchors				
	Module 080	Module 081	Module 082	Module 083	Total
North Row	5/10	3/18	0/18	3/18	11/64
Middle1 Row	9/10	(Failure)	11/20	8/20	28/50*
Middle2 Row	3/10	3/20	4/20	4/20	14/70
South Row	0/10	3/18	1/18	0/18	4/64
Total	17/40	9/56	16/76	15/76	57/248

*Does not include the 20 anchors from the failure area.

HOV Tunnel	Number of Displaced Anchors/Total Anchors				
	Module 078	Module 079	Module 080	Module 081	Total
North Row	0/6	5/18	2/18	2/18	9/60
Middle Row	0/8	7/20	4/20	0/20	11/68
South Row	1/6	3/18	2/18	0/18	6/60
Total	1/20	15/56	8/56	2/56	26/188

Weight Calculations

To assist in a statistical analysis of the anchor displacement data, the dead weight carried by each row of anchors was estimated using the weights of the individual components hung from the anchors. The following table shows the weights of individual components³.

Concrete ceiling component	Weight
12' x 8' x 4" concrete ceiling panel, without port	4,865 pounds
12' x 8' x 4" concrete ceiling panel, with port	4,640 pounds
6' x 8' x 4" concrete ceiling panel	2,483 pounds
Plate $\frac{3}{4}$ x 12, L 8 x 6 x $\frac{1}{2}$ (2) panel support	3,200 pounds
WT 13.5 x 42 panel support	1,800 pounds
2 anchor hanger support connection	46 pounds
4 anchor hanger support connection	163 pounds
WT5 ceiling strut	varied
1 panel ventilation system	50 pounds

The weight supported by the middle row of anchors in one module of the eastbound tunnel would have included

half the weight of four of the 12-foot by 8-foot panels without exhaust port	9,730 pounds,
half the weight of six of the 12-foot by 8-foot panels with exhaust port	13,920 pounds,
the full weight of one of the steel plate and "L" bracket panel supports	3,200 pounds,
the weight of six of the 2-anchor hanger support connections	276 pounds,
and the weight of two of the 4-anchor hanger support connections	326 pounds,
for a total of	27,452 pounds.

The weight supported by the north row of anchors in one module of the east-bound tunnel would have included

half the weight of two of the 12-foot by 8-foot panels without exhaust port	4,865 pounds,
half the weight of three of the 12-foot by 8-foot panels with exhaust port	6,960 pounds,
the full weight of a WT panel support	1,800 pounds,
the weight of seven of the 2-anchor hanger support connections	322 pounds,
the weight of one of the 4-anchor hanger support connections	163 pounds,
the weight of 20 of the ventilation system panels	1,000 pounds,
and half the weight of five of the WT5 ceiling struts	372 pounds,
for a total of	15,482 pounds.

³ The weight of a concrete ceiling panel, both with and without the exhaust port, was measured using a load cell provided by FHWA. The weights of the built-up WT5 steel ceiling struts were calculated using a standard cross section for beam elements and measurements of the length of the element. The weights of the hanger support connection assemblies were measured by Dan Walsh, NTSB Highway Engineer. The 2-anchor hanger support connection consists of the ceiling fitting, the rod hanger with turnbuckle, and the pins used to secure the hanger at each end. The 4-anchor hanger support connection consists of the ceiling fitting, one vertical and two 45-degree rod hangers with turnbuckles, and the pins used to secure the hangers at each end.

The weight supported by the west-bound middle row of anchors was the same as the east-bound middle1 row of anchors (27,477 pounds). The weight supported by the HOV north, west-bound north, and west-bound south rows of anchors was the same as the east-bound north row of anchors (15,647 pounds), with the exception of relatively small changes from differences in the length of the WT5 ceiling struts. Using similar calculations for the other rows for the three tunnels⁴ generated the following approximations for the dead weight supported by each row in a module. Also shown for each row is the percent of the maximum weight supported.

West-bound north row	15,647 pounds	57 percent
West-bound middle row	27,452 pounds	100 percent
West-bound south row	15,440 pounds	56 percent
East-bound north row	15,482 pounds	56 percent
East-bound middle1 row	27,452 pounds	100 percent
East bound middle2 row	22,834 pounds	83 percent
East-bound south row	9,617 pounds	35 percent
HOV north row	15,456 pounds	56 percent
HOV middle row	21,891 pounds	80 percent
HOV south row	9,699 pounds	35 percent

James F. Wildey II
Supervisory Metallurgist

⁴ The HOV tunnel ventilation duct was constructed using 12-foot wide concrete panels between the north row and middle row of anchors and 6-foot wide concrete panels between the middle row and south row of anchors. The west-bound tunnel ventilation duct was constructed using 12-foot wide concrete panels between the north row and middle row of anchors and 12-foot wide concrete panels between the middle row and south row of anchors. Not included in any of these calculations is the weight of the 12 inch wide by 3/8 inch thick steel plates that span the gap between modules.

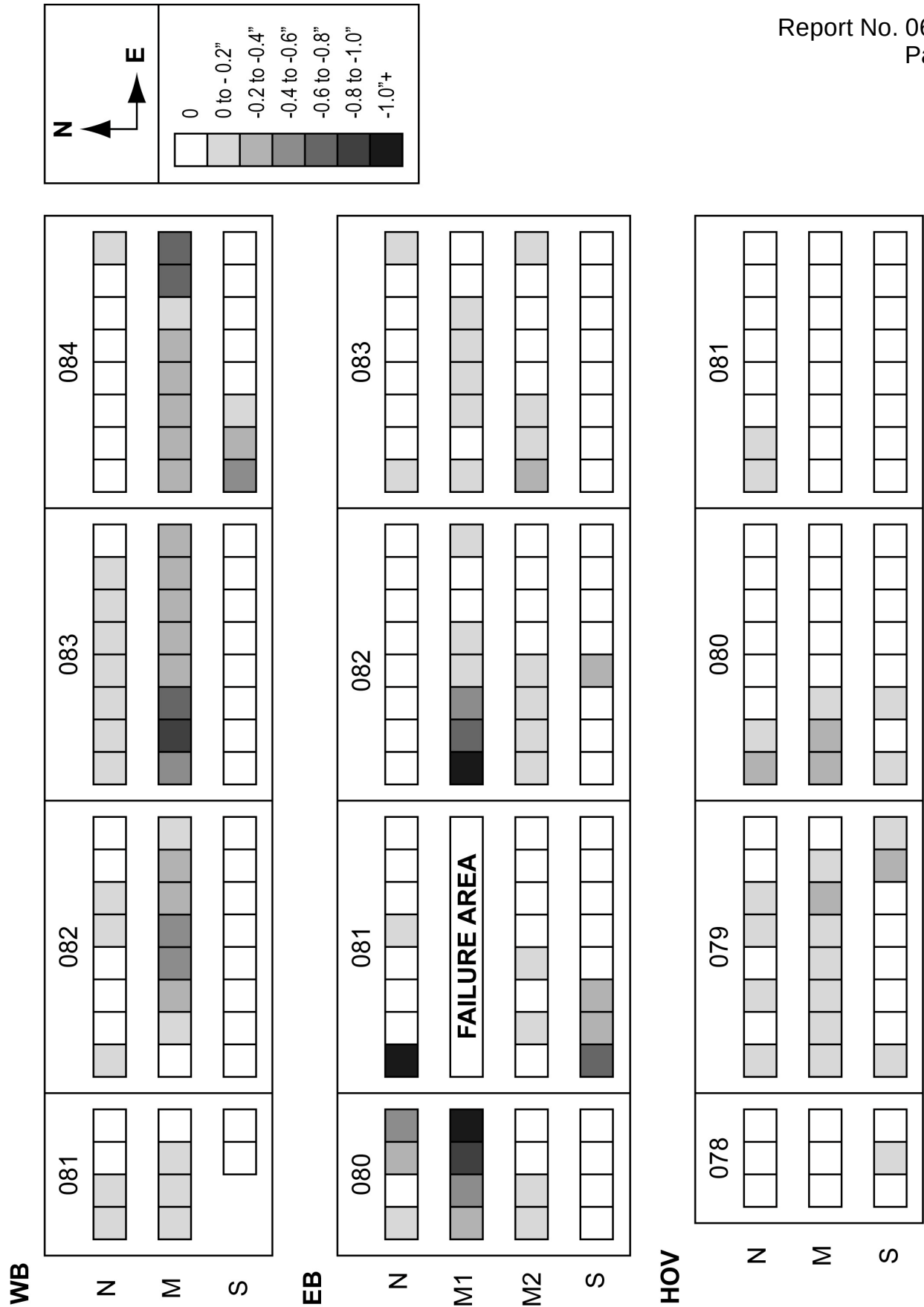
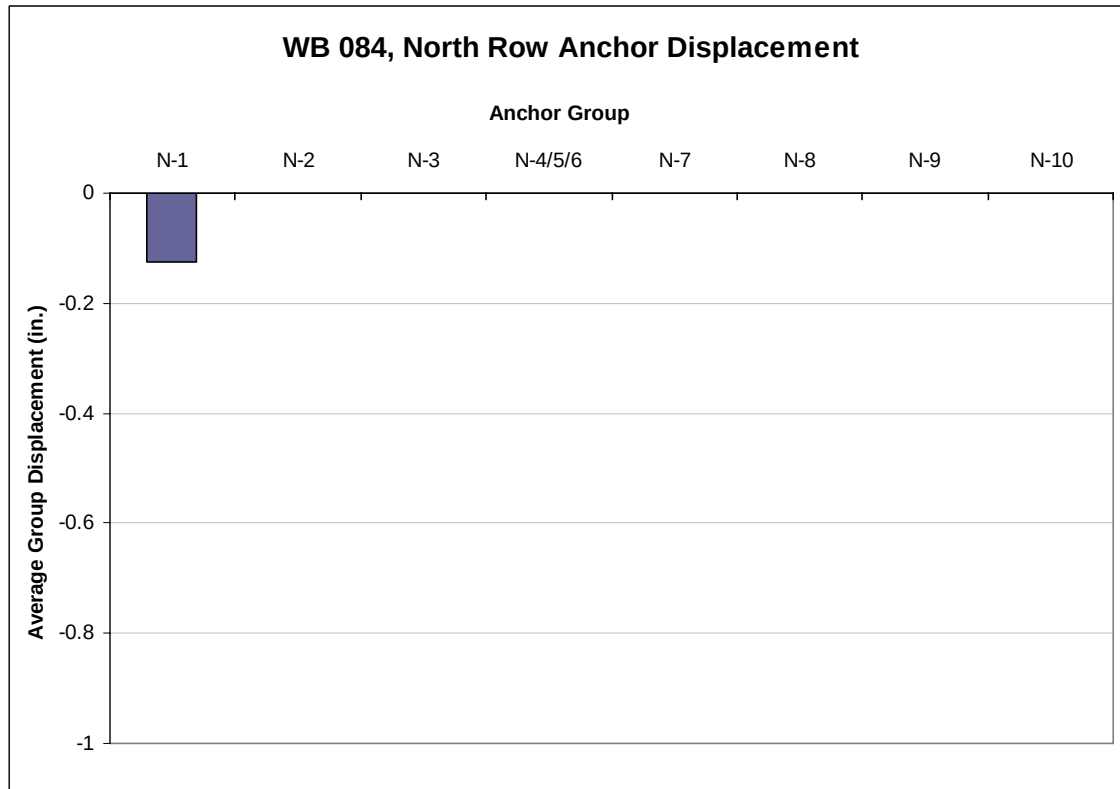
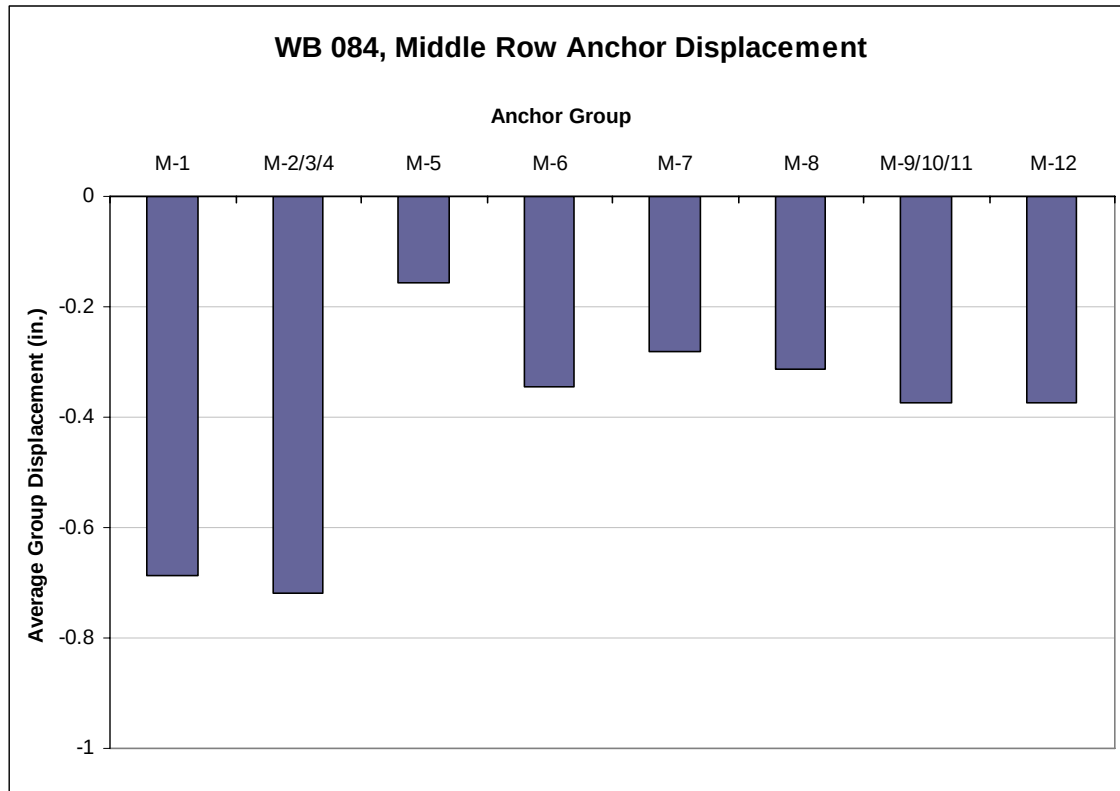


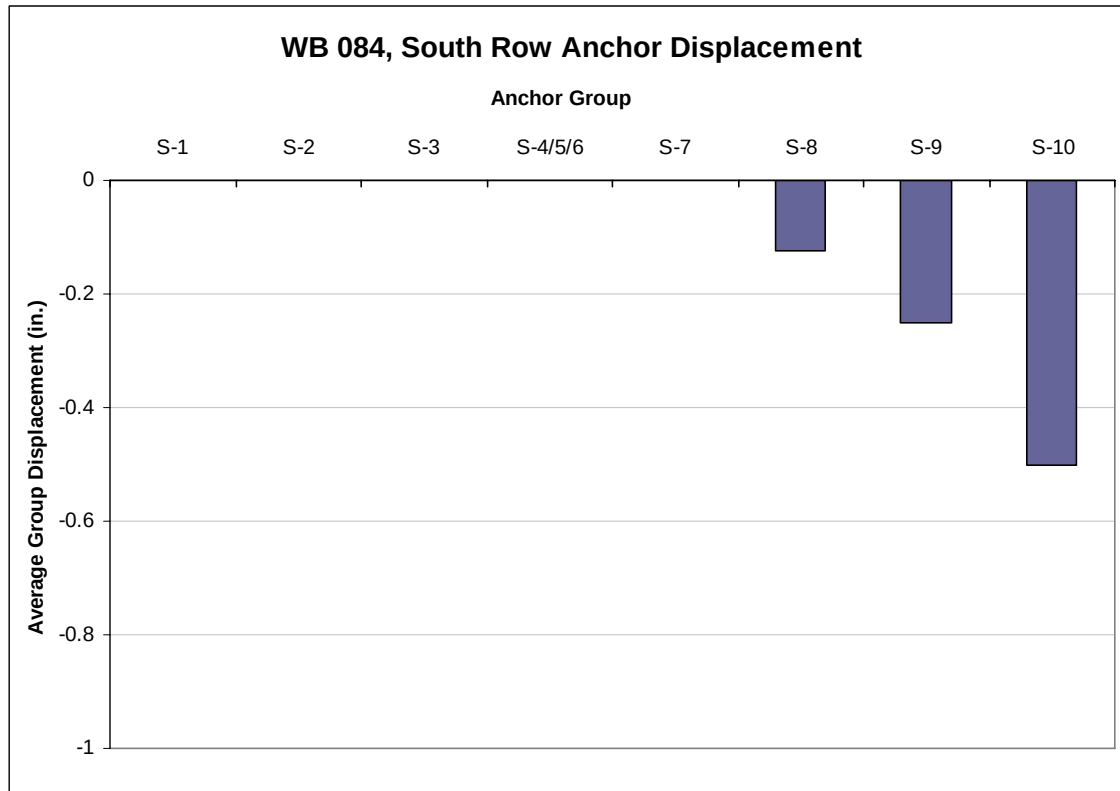
Figure 1. Graphical representation of the downward displacement of the ceiling fittings in each module of the WB, EB, and HOV tunnels.



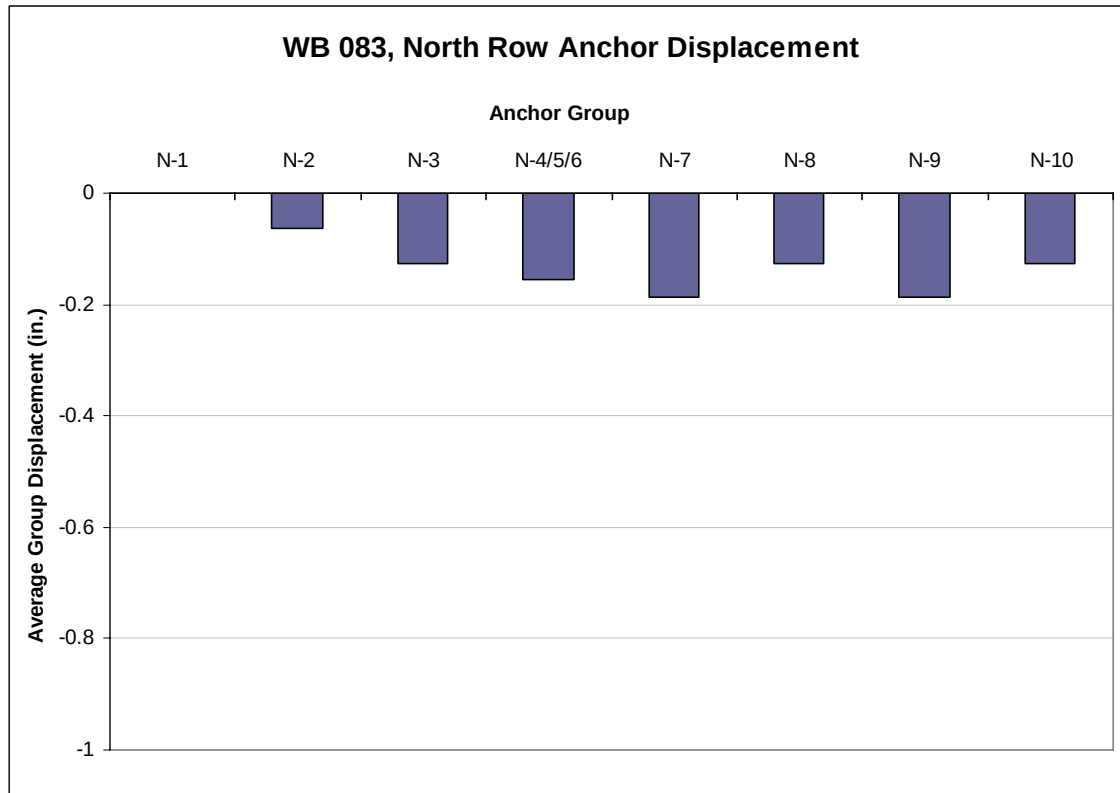
WB 084, North Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
N-1	-0.125	-0.125					-0.125
N-2	0	0					0
N-3	0	0					0
N-4/5/6			0	0	0	0	0
N-7	0	0					0
N-8	0	0					0
N-9	0	0					0
N-10	0	0					0



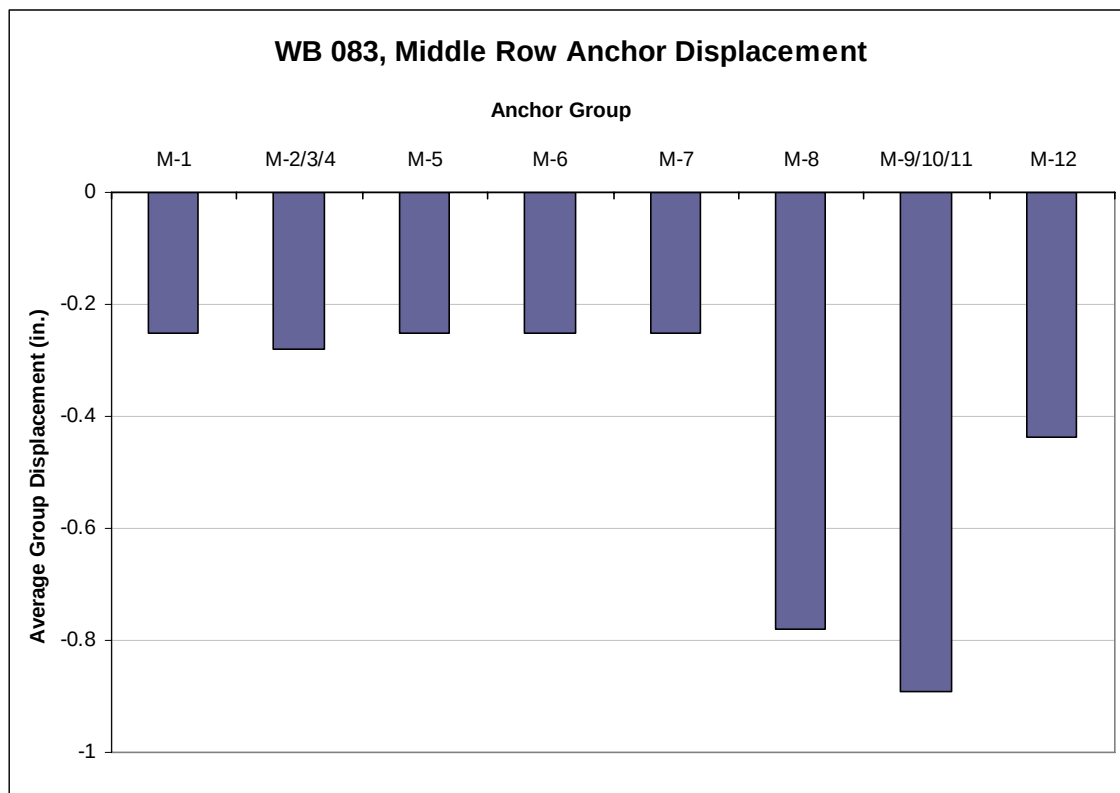
WB 084, Middle Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
M-1	-0.625	-0.75					-0.6875
M-2/3/4			-1.125	-0.25	-1.25	-0.25	-0.71875
M-5	-0.25	-0.0625					-0.15625
M-6	-0.0625	-0.625					-0.34375
M-7	-0.0625	-0.5					-0.28125
M-8	-0.625	0					-0.3125
M-9/10/11			-0.625	-0.125	-0.625	-0.125	-0.375
M-12	0	-0.75					-0.375



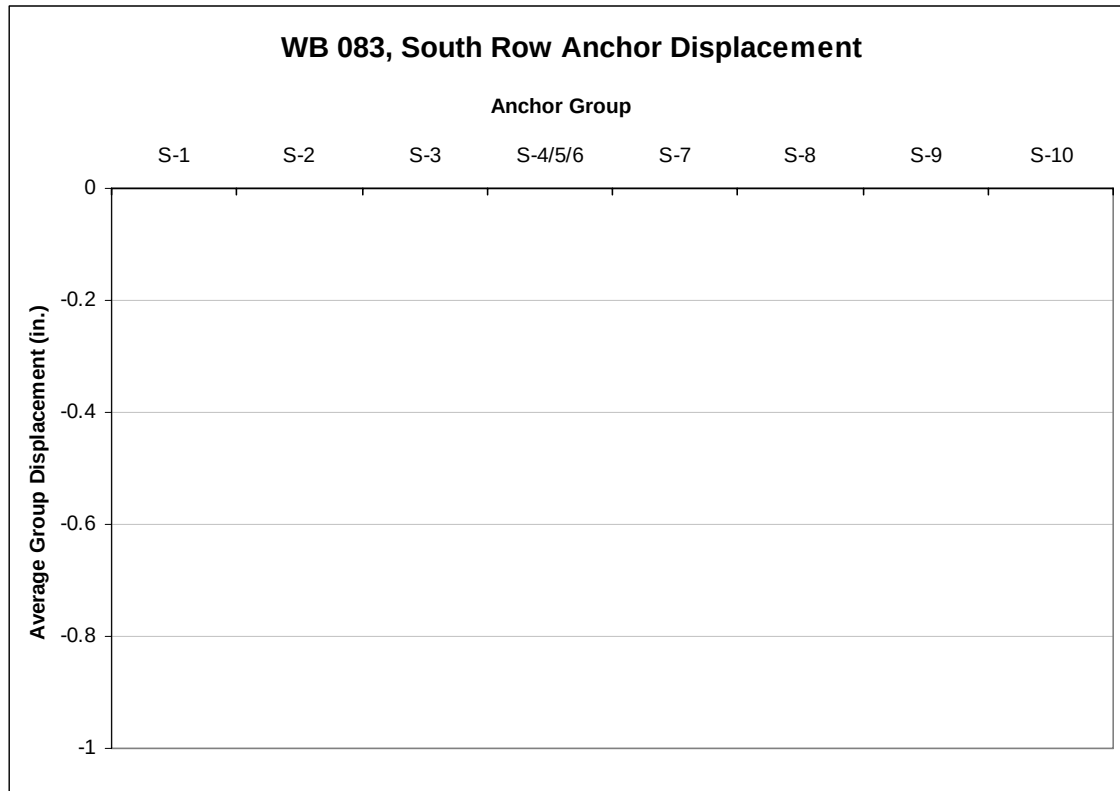
WB 084, South Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
S-1	0	0					0
S-2	0	0					0
S-3	0	0					0
S-4/5/6			0	0	0	0	0
S-7	0	0					0
S-8	0	-0.25					-0.125
S-9	-0.5	0					-0.25
S-10	-0.125	-0.875					-0.5



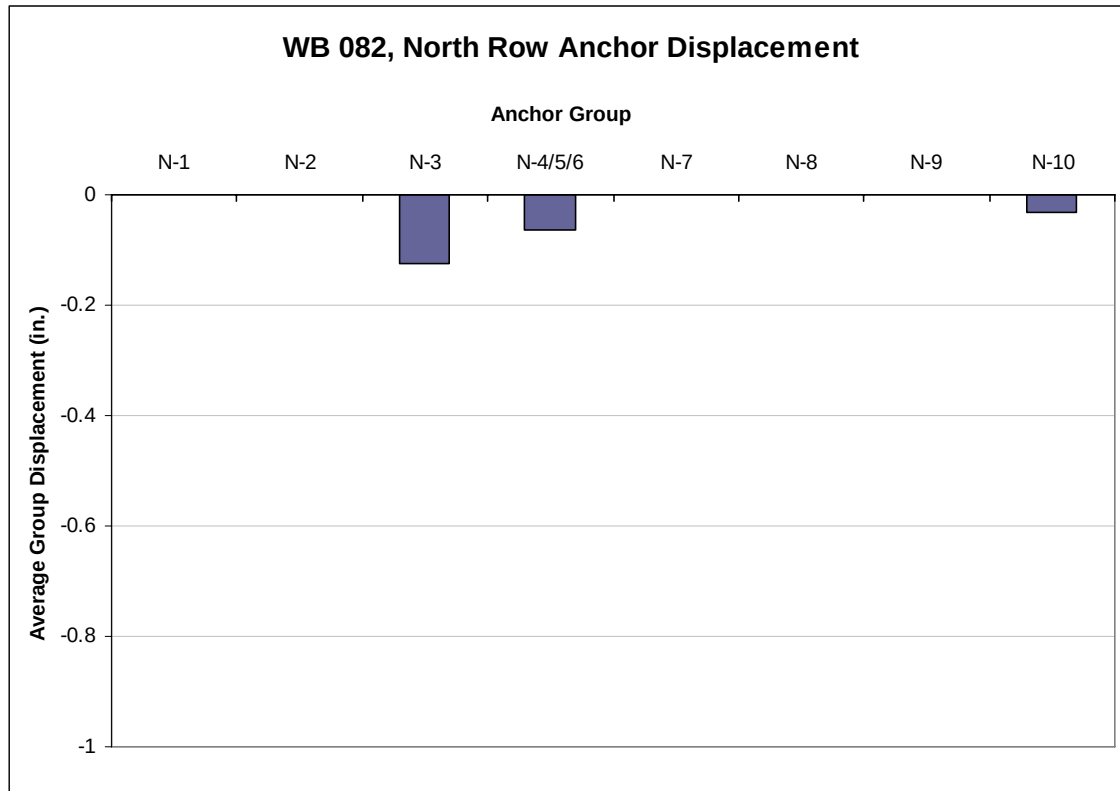
WB 083, North Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
N-1	0	0					0
N-2	-0.125	0					-0.0625
N-3	0	-0.25					-0.125
N-4/5/6			-0.25	-0.25	-0.125	0	-0.15625
N-7	0	-0.375					-0.1875
N-8	0	-0.25					-0.125
N-9	-0.375	0					-0.1875
N-10	0	-0.25					-0.125



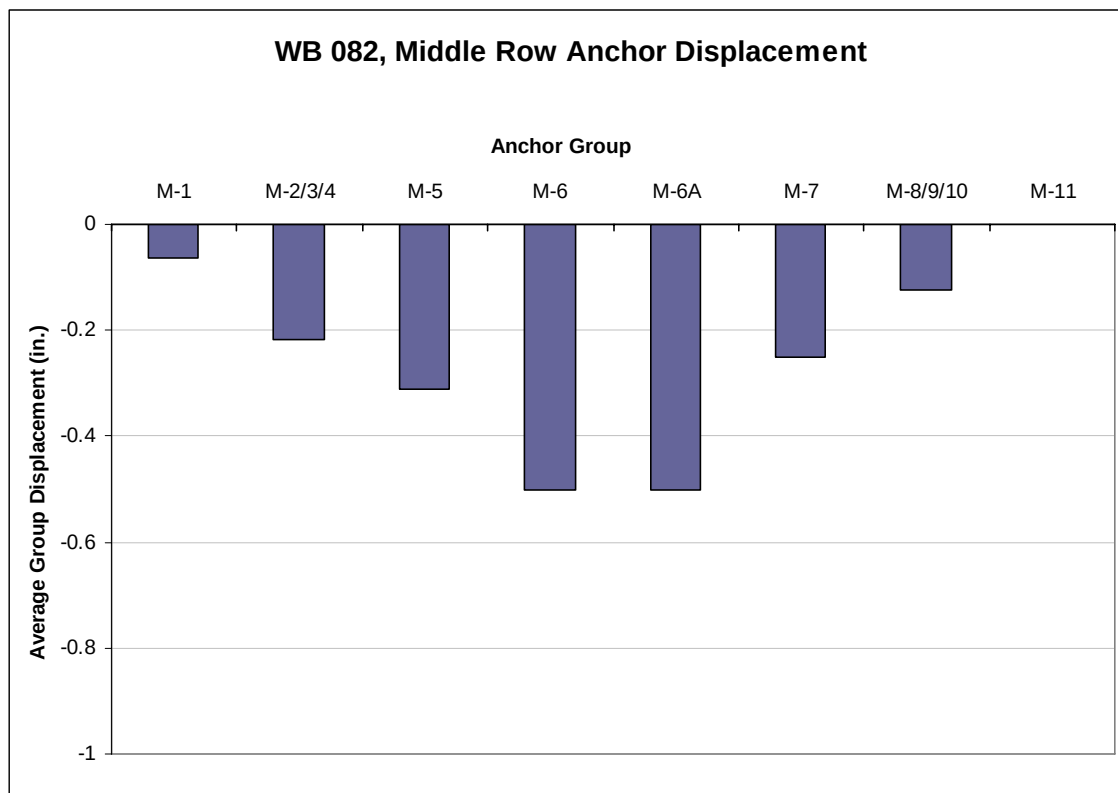
WB 083, Middle Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
M-1	-0.125	-0.375					-0.25
M-2/3/4			-0.125	-0.5	-0.125	-0.375	-0.28125
M-5	-0.125	-0.375					-0.25
M-6	0	-0.5					-0.25
M-7	0	-0.5					-0.25
M-8	-1.5	-0.0625					-0.78125
M-9/10/11			-0.125	-1.625	-0.0625	-1.75	-0.890625
M-12	-0.5	-0.375					-0.4375



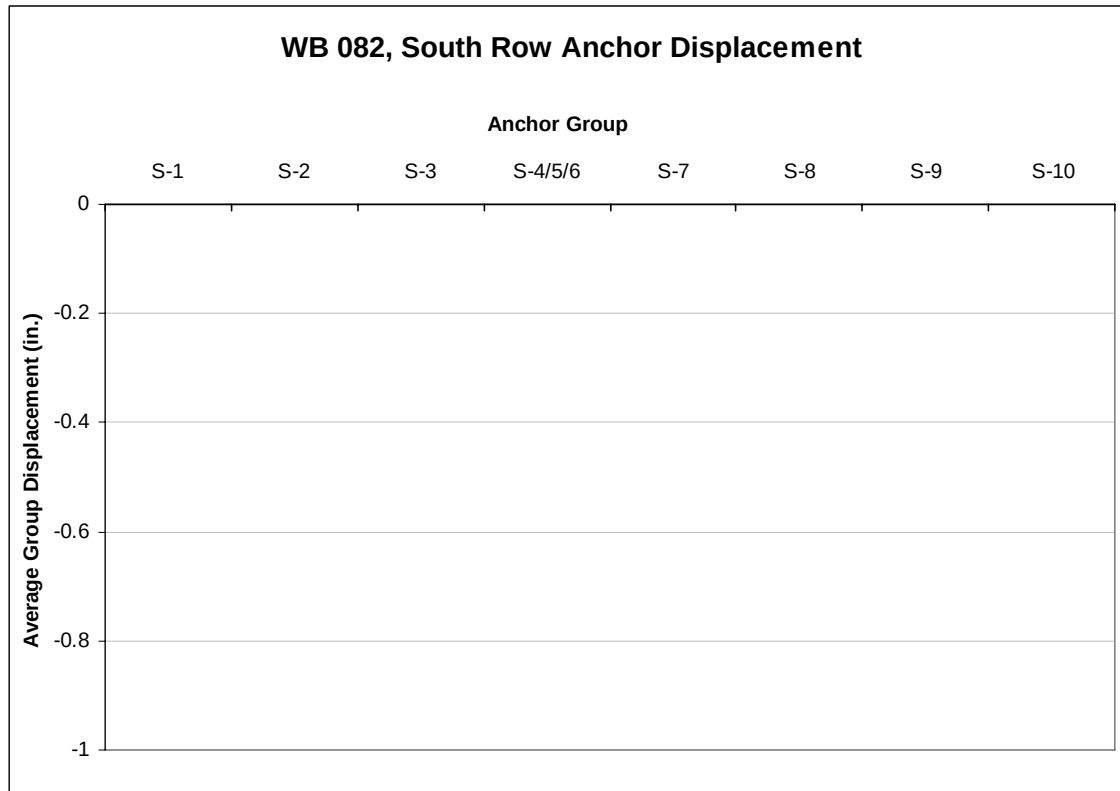
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Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
S-1	0	0					0
S-2	0	0					0
S-3	0	0					0
S-4/5/6			0	0	0	0	0
S-7	0	0					0
S-8	0	0					0
S-9	0	0					0
S-10	0	0					0



WB 082, North Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
N-1	0	0					0
N-2	0	0					0
N-3	-0.25	0					-0.125
N-4/5/6			-0.0625	-0.0625	-0.0625	-0.0625	-0.0625
N-7	0	0					0
N-8	0	0					0
N-9	0	0					0
N-10	-0.0625	0					-0.03125

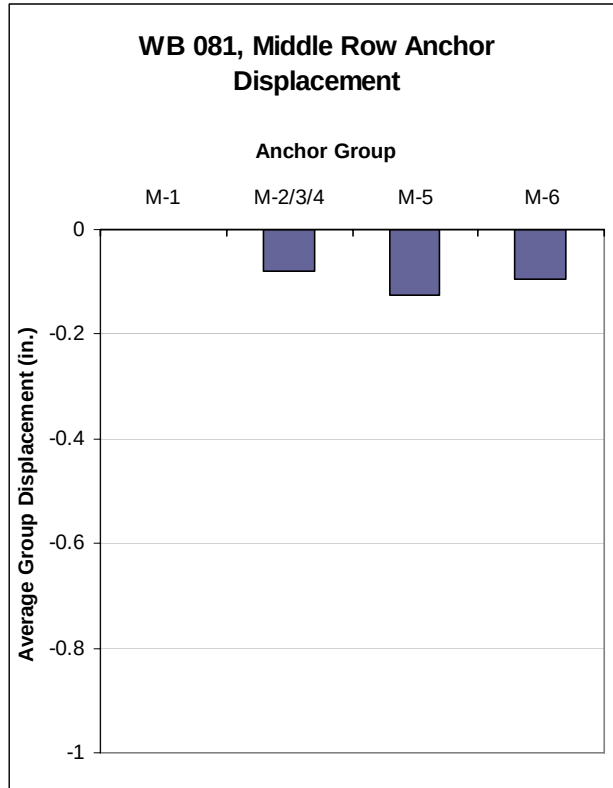


WB 082, Middle Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
M-1	0	-0.125					-0.0625
M-2/3/4			-0.125	0	-0.375	-0.375	-0.21875
M-5	0	-0.625					-0.3125
M-6	0	-1					-0.5
M-6A	0	-1					-0.5
M-7	-0.5	0					-0.25
M-8/9/10			-0.125	-0.25	0	-0.125	-0.125
M-11	0	0					0



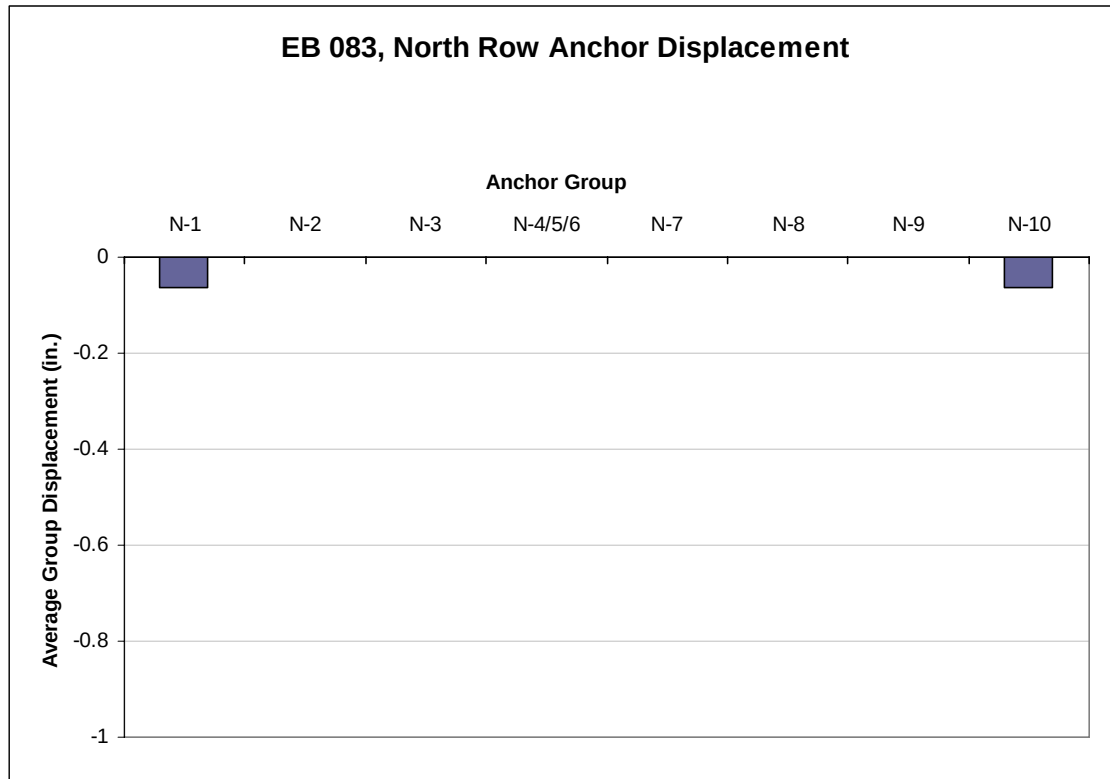
WB 082, South Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	<i>N</i>	<i>S</i>	<i>NE</i>	<i>SE</i>	<i>NW</i>	<i>SW</i>	
S-1	0	0					0
S-2	0	0					0
S-3	0	0					0
S-4/5/6			0	0	0	0	0
S-7	0	0					0
S-8	0	0					0
S-9	0	0					0
S-10	0	0					0

WB 081, North Row Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
N-1	0	0					0
N-2	0	0					0
N-3	-0.25	0					-0.125
N-4/5/6			-0.25	-0.25	0	0	-0.125

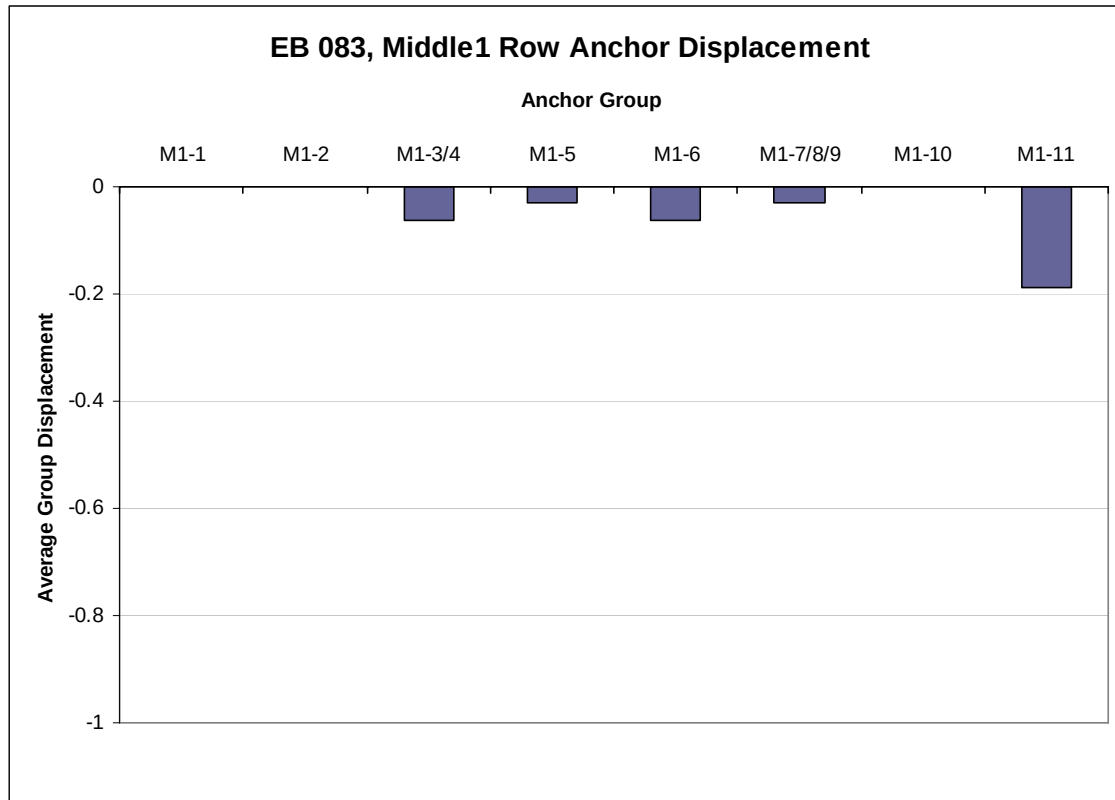


WB 081, Middle Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
M-1	0	0					0
M-2/3/4			-0.125	-0.125	0	-0.0625	-0.078125
M-5	-0.125	-0.125					-0.125
M-6	-0.125	-0.0625					-0.09375

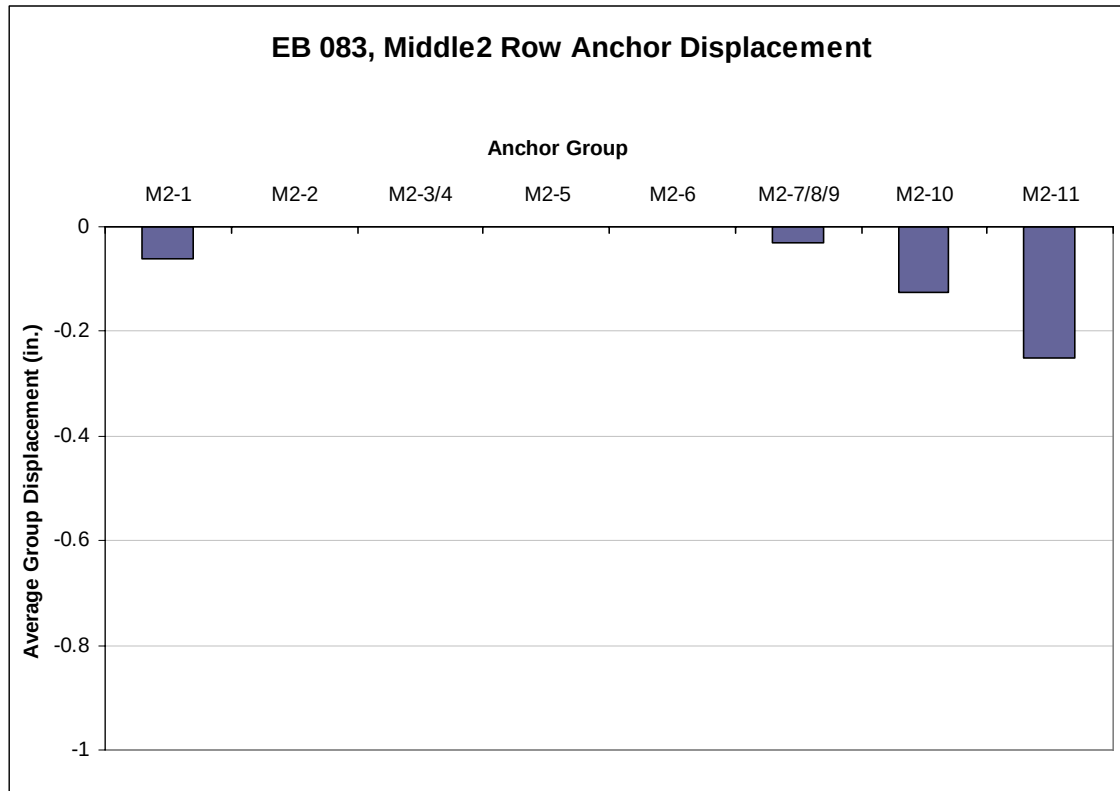
WB 081, South Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
S-1	0	0					0
S-2	0	0					0



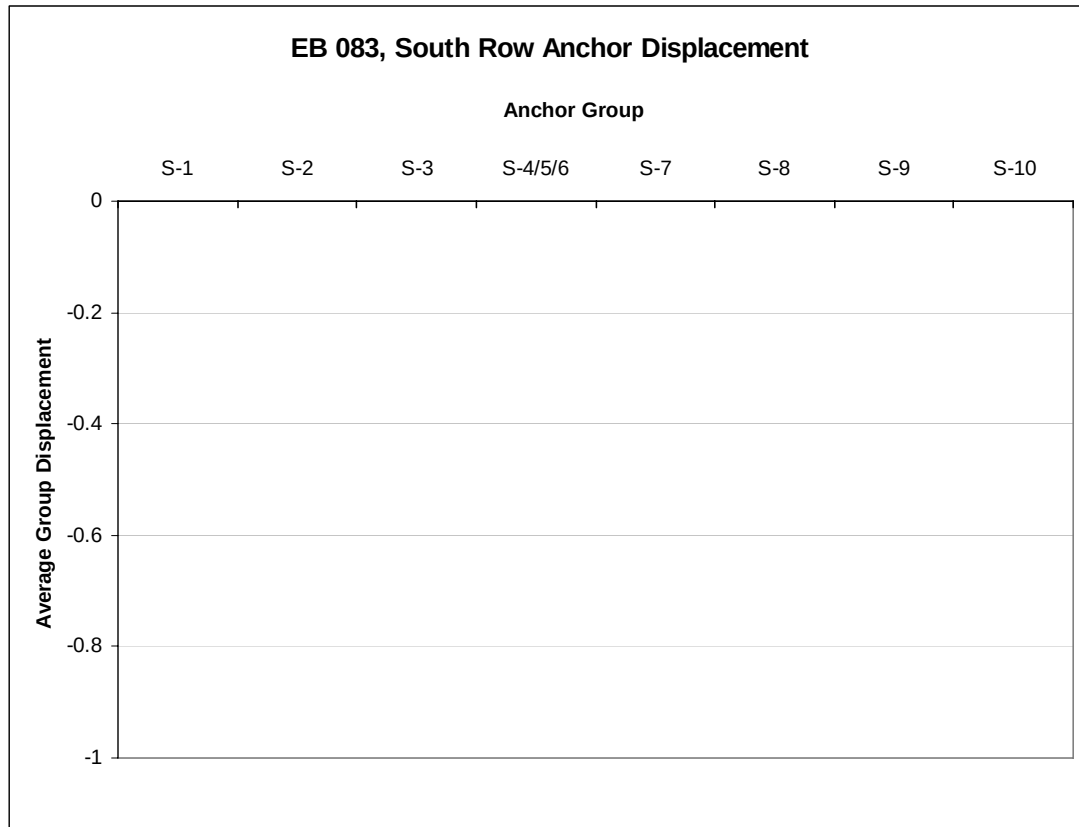
EB 083, North Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
N-1	0	-0.125					-0.0625
N-2	0	0					0
N-3	0	0					0
N-4/5/6			0	0	0	0	0
N-7	0	0					0
N-8	0	0					0
N-9	0	0					0
N-10	-0.0625	-0.0625					-0.0625



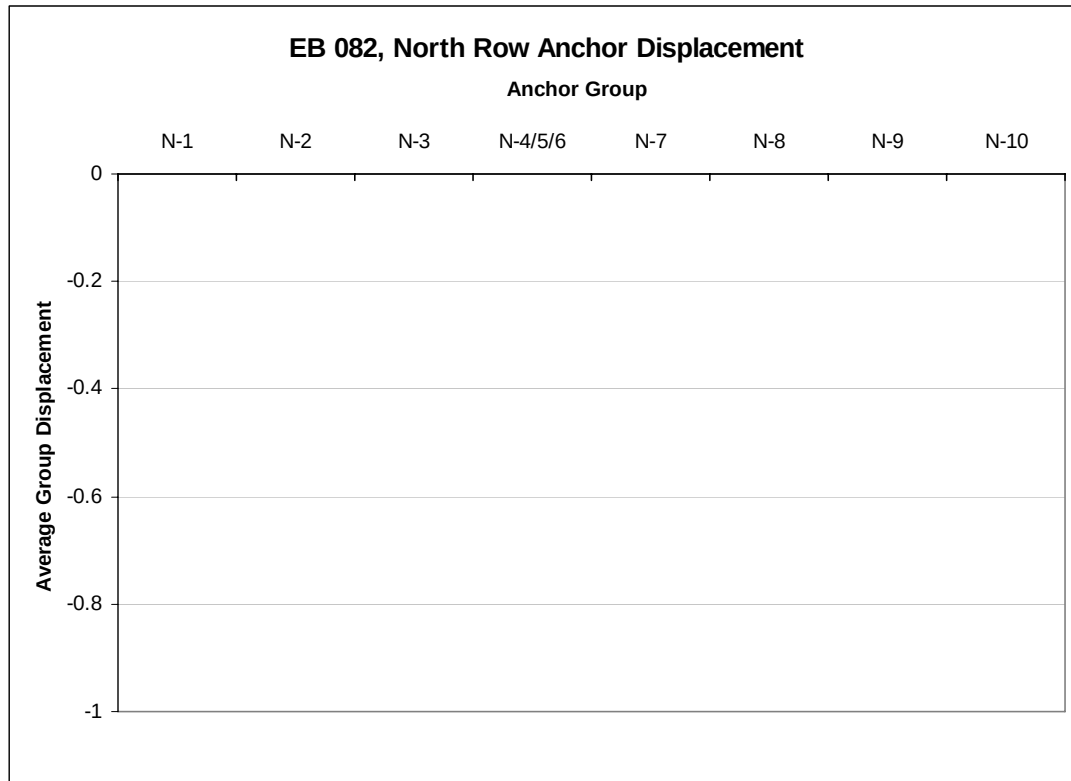
EB 083, Middle1 Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
M1-1	0	0					0
M1-2	0	0					0
M1-3/4			0	-0.125	0	-0.125	-0.0625
M1-5	0	-0.0625					-0.03125
M1-6	0	-0.125					-0.0625
M1-7/8/9			0	0	-0.0625	-0.0625	-0.03125
M1-10	0	0					0
M1-11	-0.25	-0.125					-0.1875



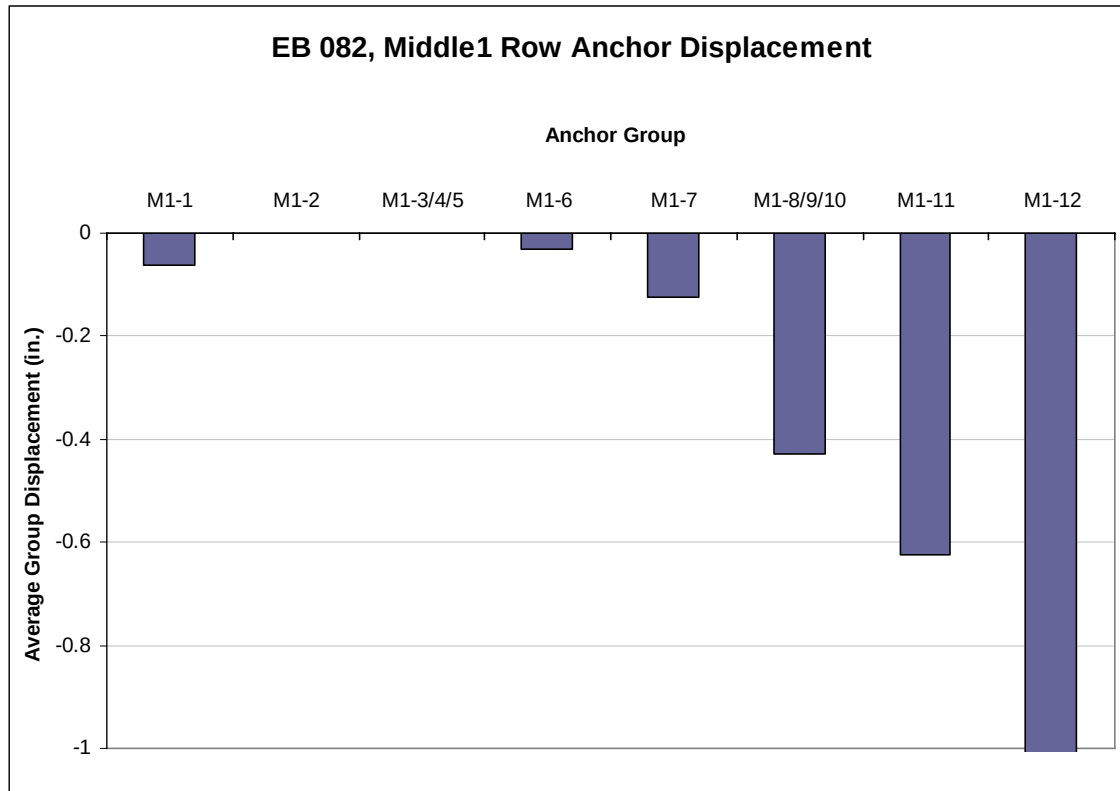
EB 083, Middle2 Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
M2-1	-0.125	0					-0.0625
M2-2	0	0					0
M2-3/4			0	0	0	0	0
M2-5	0	0					0
M2-6	0	0					0
M2-7/8/9			0	0	0	-0.125	-0.03125
M2-10	-0.25	0					-0.125
M2-11	-0.5	0					-0.25



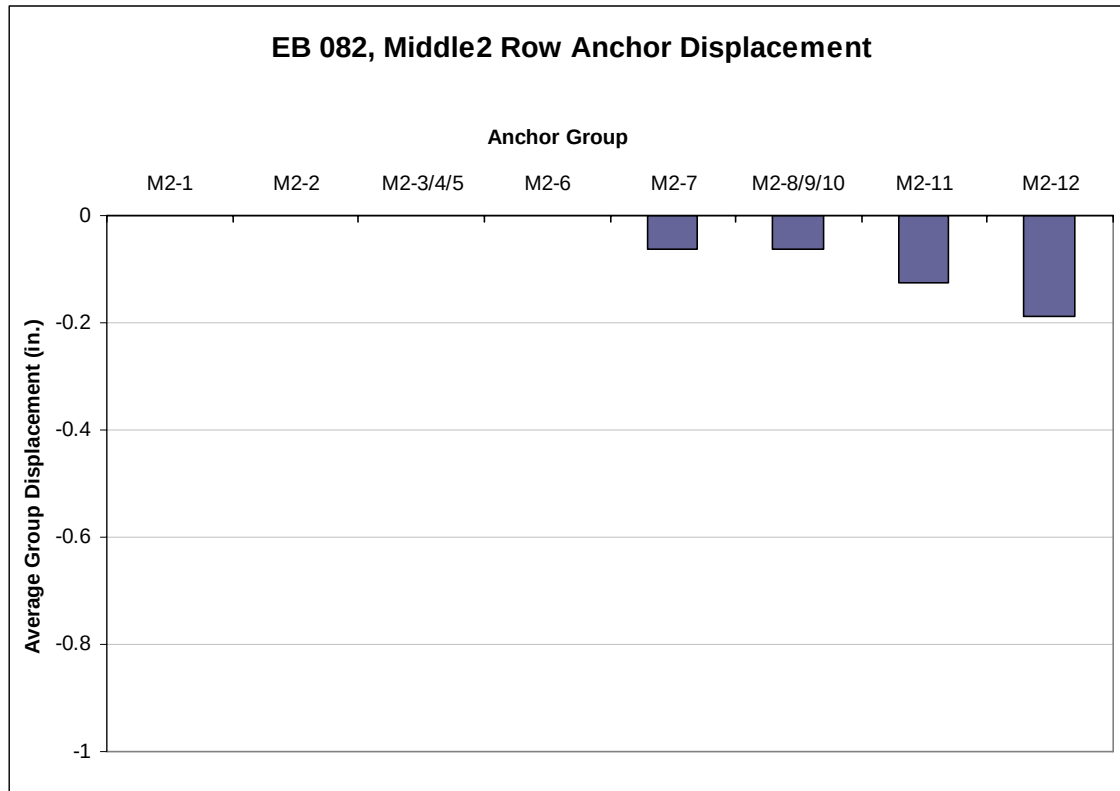
EB 083, South Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
S-1	0	0					0
S-2	0	0					0
S-3	0	0					0
S-4/5/6			0	0	0	0	0
S-7	0	0					0
S-8	0	0					0
S-9	0	0					0
S-10	0	0					0



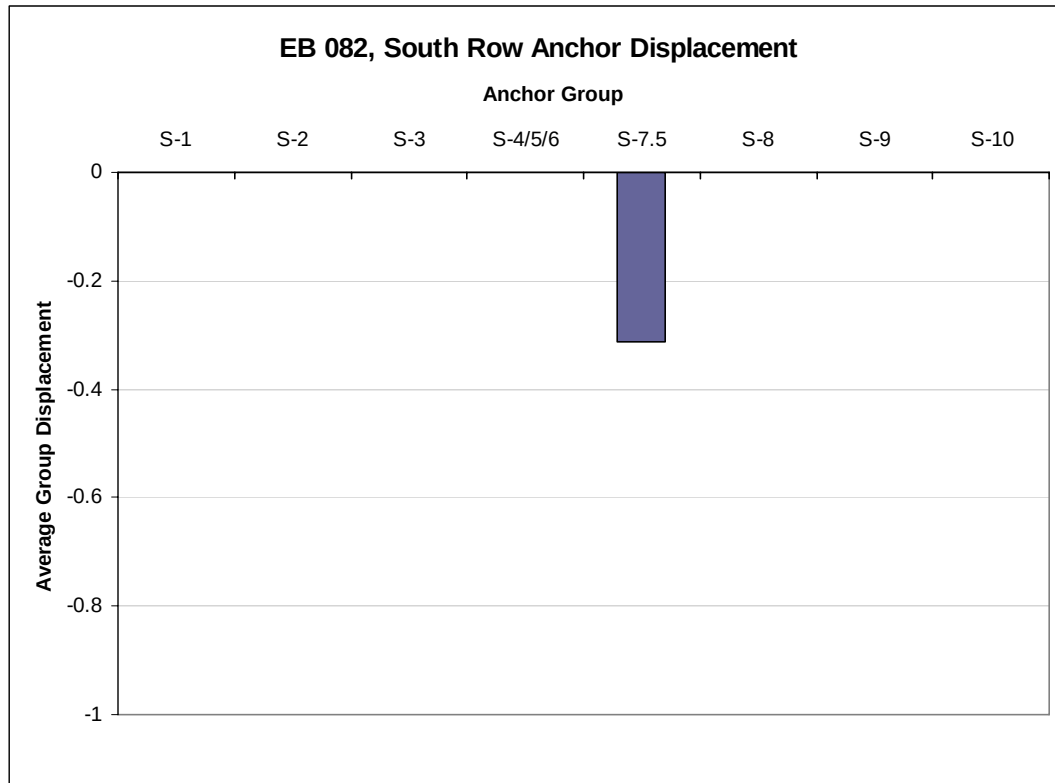
EB 082, North Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
N-1	0	0					0
N-2	0	0					0
N-3	0	0					0
N-4/5/6			0	0	0	0	0
N-7	0	0					0
N-8	0	0					0
N-9	0	0					0
N-10	0	0					0



EB 082, Middle1 Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
M1-1	0	-0.125					-0.0625
M1-2	0	0					0
M1-3/4/5			0	0	0	0	0
M1-6	-0.0625	0					-0.03125
M1-7	-0.125	-0.125					-0.125
M1-8/9/10			-0.34	-0.17	-0.61	-0.6	-0.43
M1-11	0	-1.25					-0.625
M1-12	-0.34	-2.5					-1.42

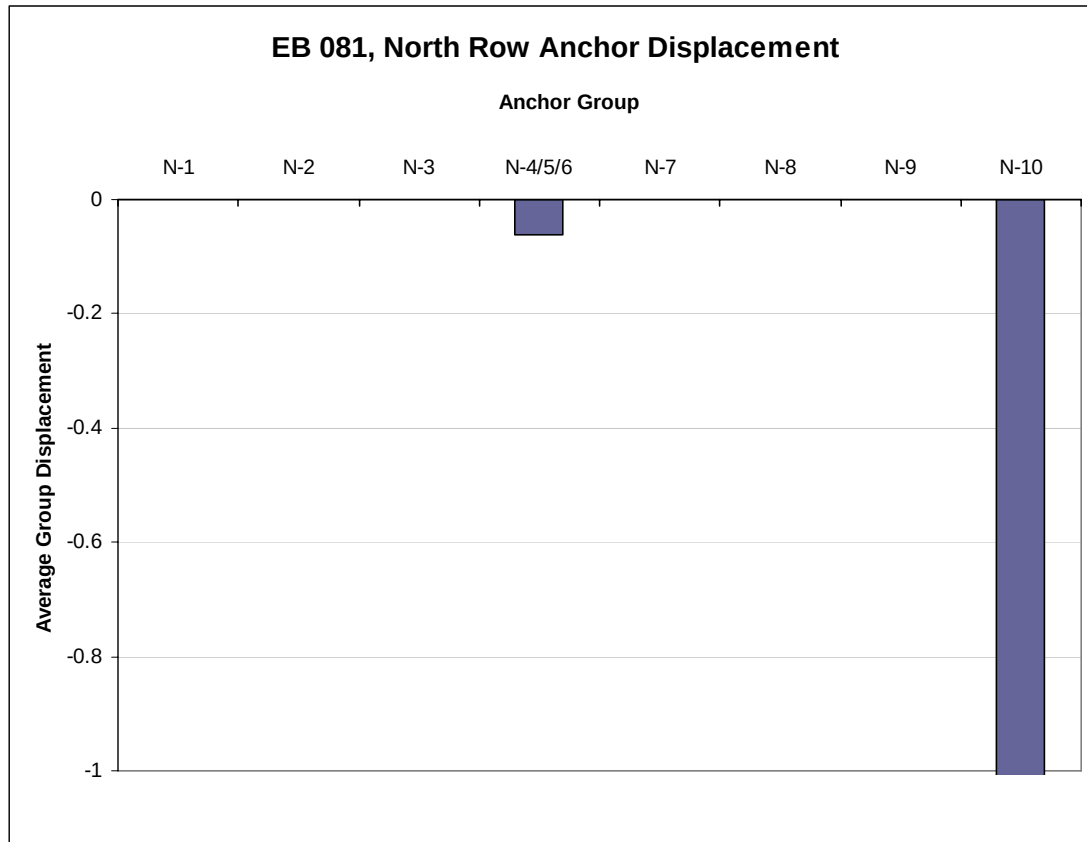


EB 082, Middle2 Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
M2-1	0	0					0
M2-2	0	0					0
M2-3/4/5			0	0	0	0	0
M2-6	0	0					0
M2-7	-0.125	0					-0.0625
M2-8/9/10			0	0	-0.25	0	-0.0625
M2-11	0	-0.25					-0.125
M2-12	-0.375	0					-0.1875

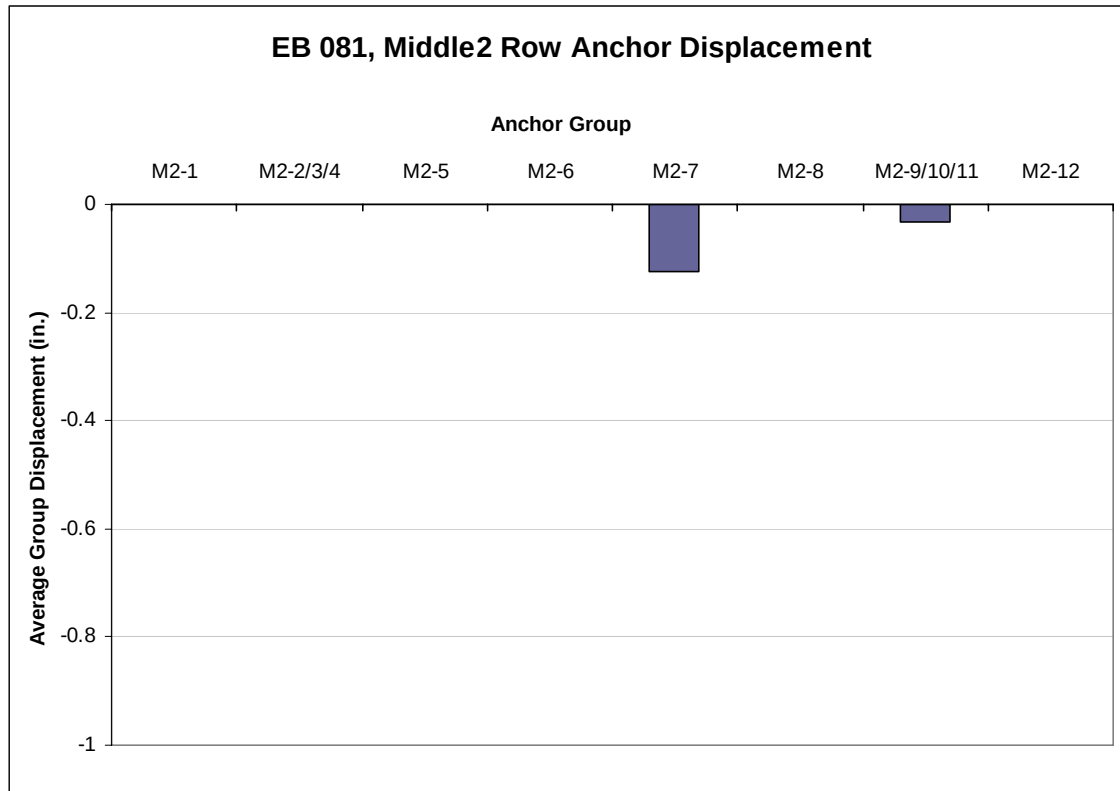


EB 082, South Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
S-1	0	0					0
S-2	0	0					0
S-3	0	0					0
S-4/5/6			0	0	0	0	0
S-7.5	0	-0.625					-0.3125
S-8	0	0					0
S-9	0	0					0
S-10	0	0					0

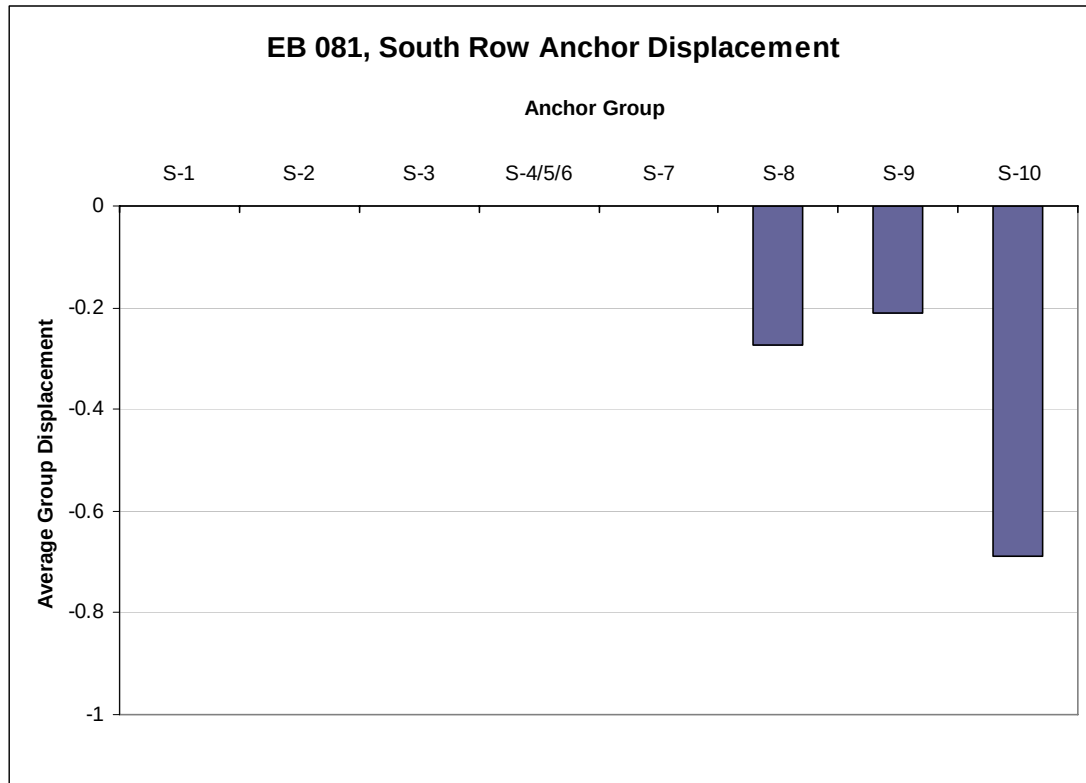
Note: During the on-scene documentation by the NTSB, two pairs of anchors were found at the EB082-S-7 position, one of which was labeled “7.5” and one of which was labeled “7”. Displacement was found in one of the “7.5” anchors, as shown in the table; therefore, the anchors in the “7” position, which were not displaced, are not included in the data. The “Removal of Hardware Log” generated by law enforcement personnel before the arrival of the NTSB, lists only one pair of anchors at the “7” position and notes that displacement was found in one of the anchors.



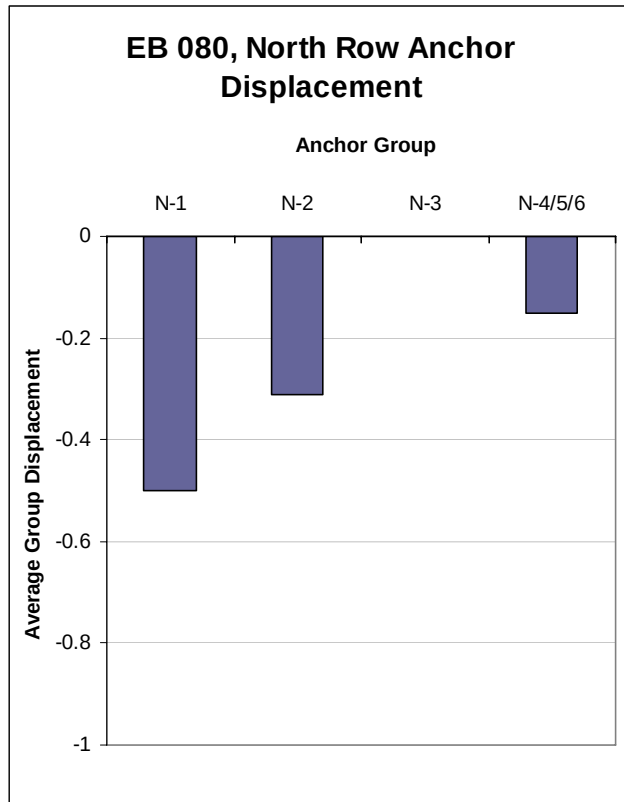
EB 081, North Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
N-1	0	0					0
N-2	0	0					0
N-3	0	0					0
N-4/5/6			-0.125	0	-0.125	0	-0.0625
N-7	0	0					0
N-8	0	0					0
N-9	0	0					0
N-10	-2.23	0					-1.115



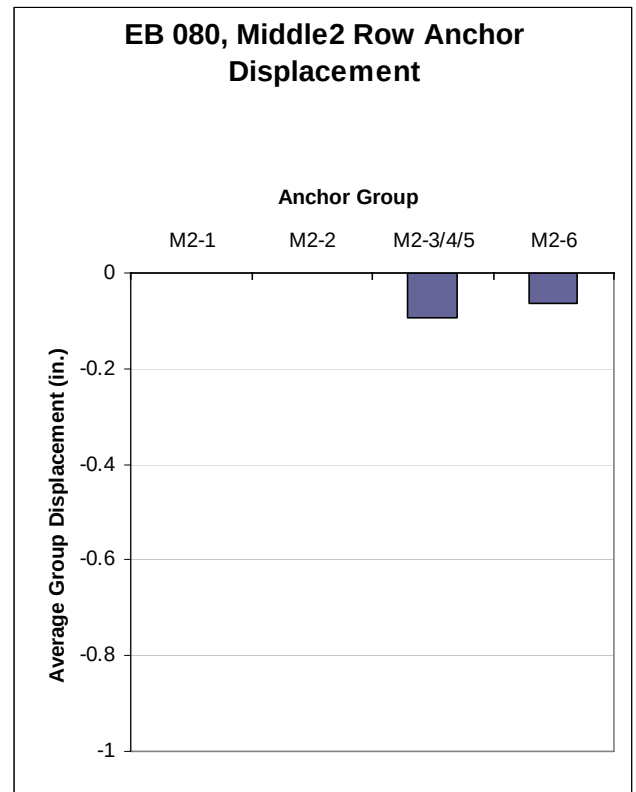
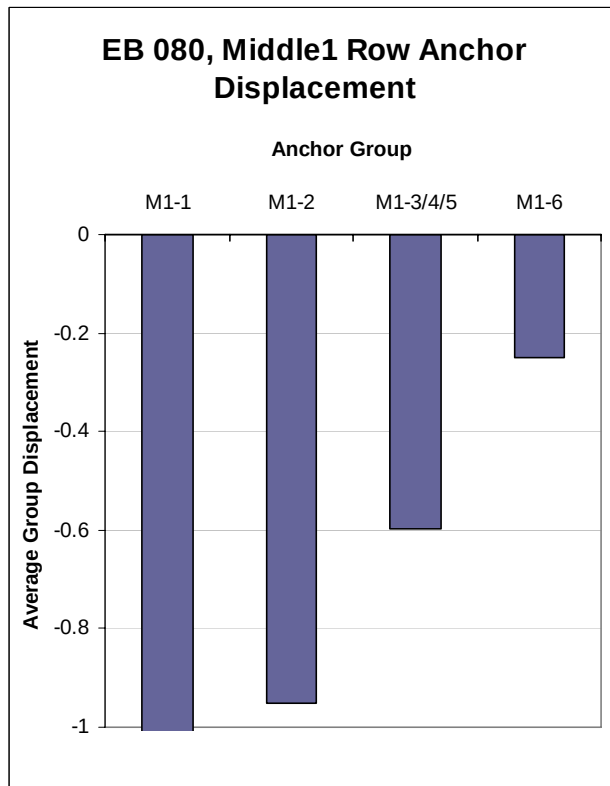
EB 081, Middle2 Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
M2-1	0	0					0
M2-2/3/4			0	0	0	0	0
M2-5	0	0					0
M2-6	0	0					0
M2-7	-0.125	-0.125					-0.125
M2-8	0	0					0
M2-9/10/11			0	-0.125	0	0	-0.03125
M2-12	0	0					0



EB 081, South Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
S-1	0	0					0
S-2	0	0					0
S-3	0	0					0
S-4/5/6			0	0	0	0	0
S-7	0	0					0
S-8	0	-0.55					-0.275
S-9	0	-0.42					-0.21
S-10	0	-1.375					-0.6875

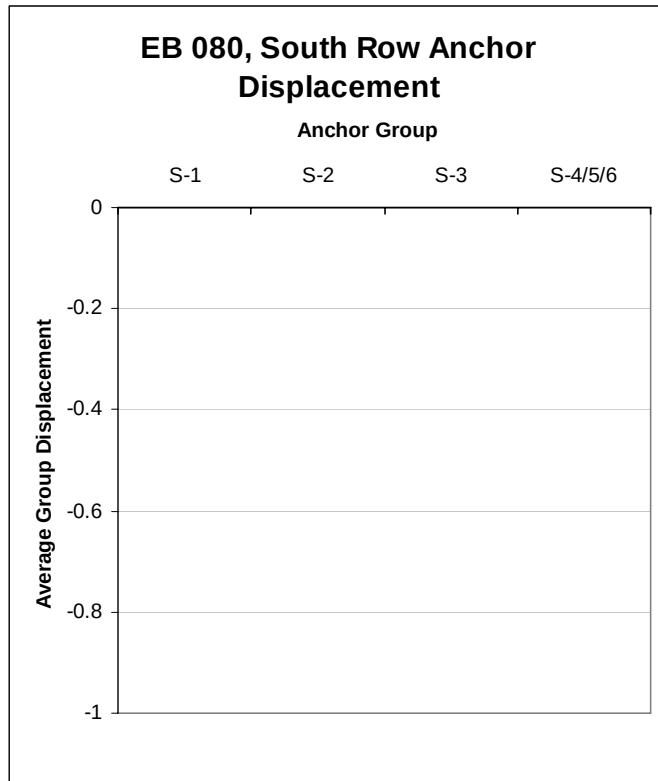


EB 080, North Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
N-1	-1	0					-0.5
N-2	-0.5	-0.125					-0.3125
N-3	0	0					0
N-4/5/6			-0.3	-0.31	0	0	-0.1525



EB 080, Middle1 Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
M1-1	-0.47	-2.65					-1.56
M1-2	-1.9	0					-0.95
M1-3/4/5			-0.875	-0.44	-0.75	-0.33	-0.59875
M1-6	-0.125	-0.375					-0.25

EB 080, Middle2 Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
M2-1	0	0					0
M2-2	0	0					0
M2-3/4/5			0	-0.25	0	-0.125	-0.09375
M2-6	0	-0.125					-0.0625

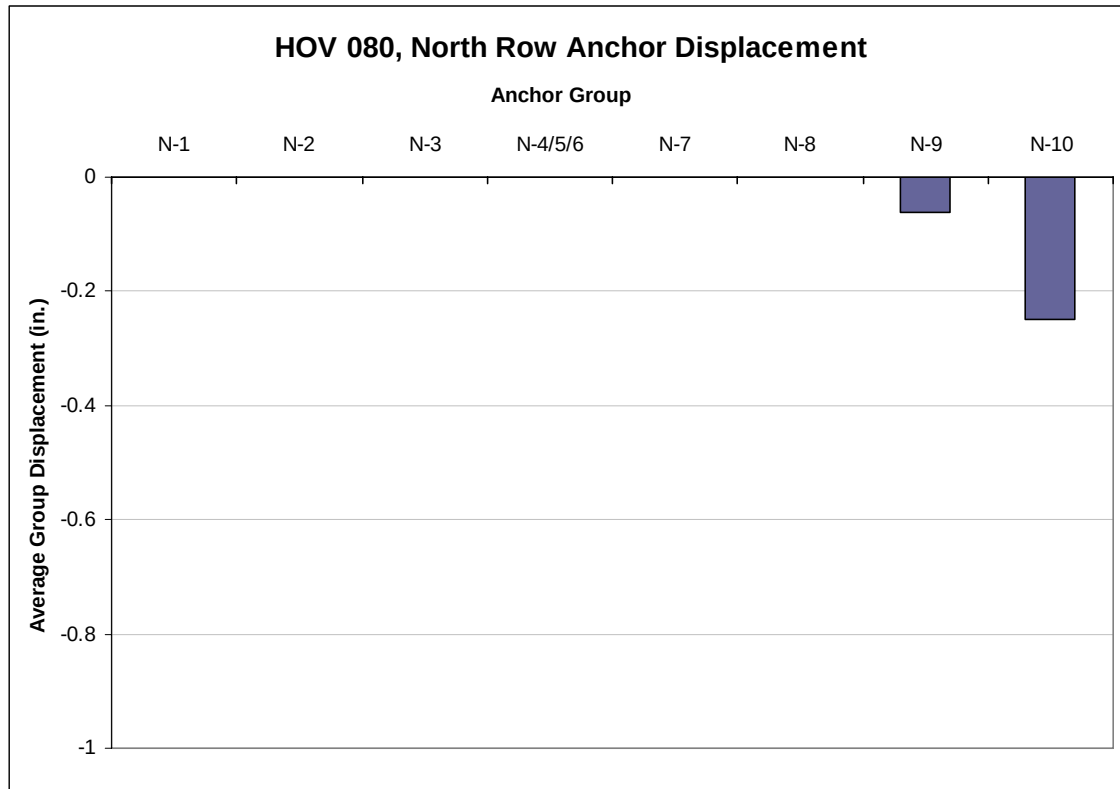


EB 080, South Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
S-1	0	0					0
S-2	0	0					0
S-3	0	0					0
S-4/5/6			0	0	0	0	0

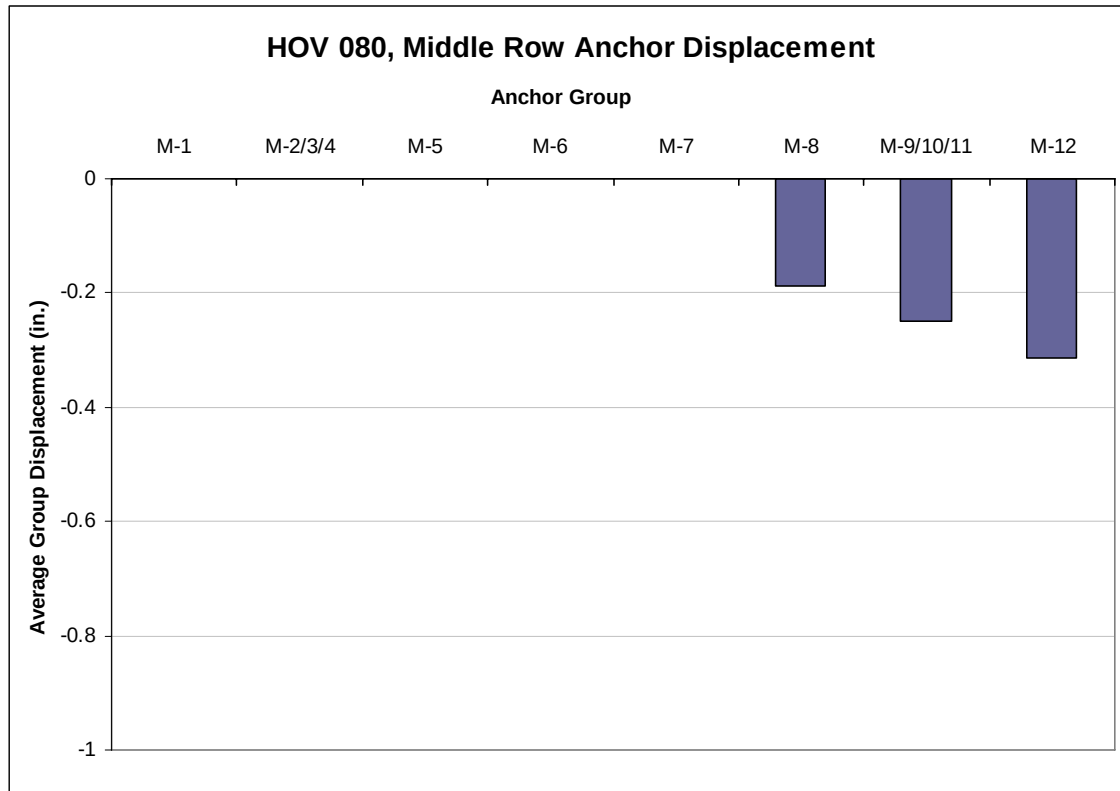
HOV 081, North Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	<i>N</i>	<i>S</i>	<i>NE</i>	<i>SE</i>	<i>NW</i>	<i>SW</i>	
N-1	0	0					0
N-2	0	0					0
N-3	0	0					0
N-4/5/6			0	0	0	0	0
N-7	0	0					0
N-8	0	0					0
N-9	-0.125	0					-0.0625
N-10	-0.125	0					-0.0625

HOV 081, Middle Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	<i>N</i>	<i>S</i>	<i>NE</i>	<i>SE</i>	<i>NW</i>	<i>SW</i>	
M-1	0	0					0
M-2/3/4			0	0	0	0	0
M-5	0	0					0
M-6	0	0					0
M-7	0	0					0
M-8	0	0					0
M-9/10/11			0	0	0	0	0
M-12	0	0					0

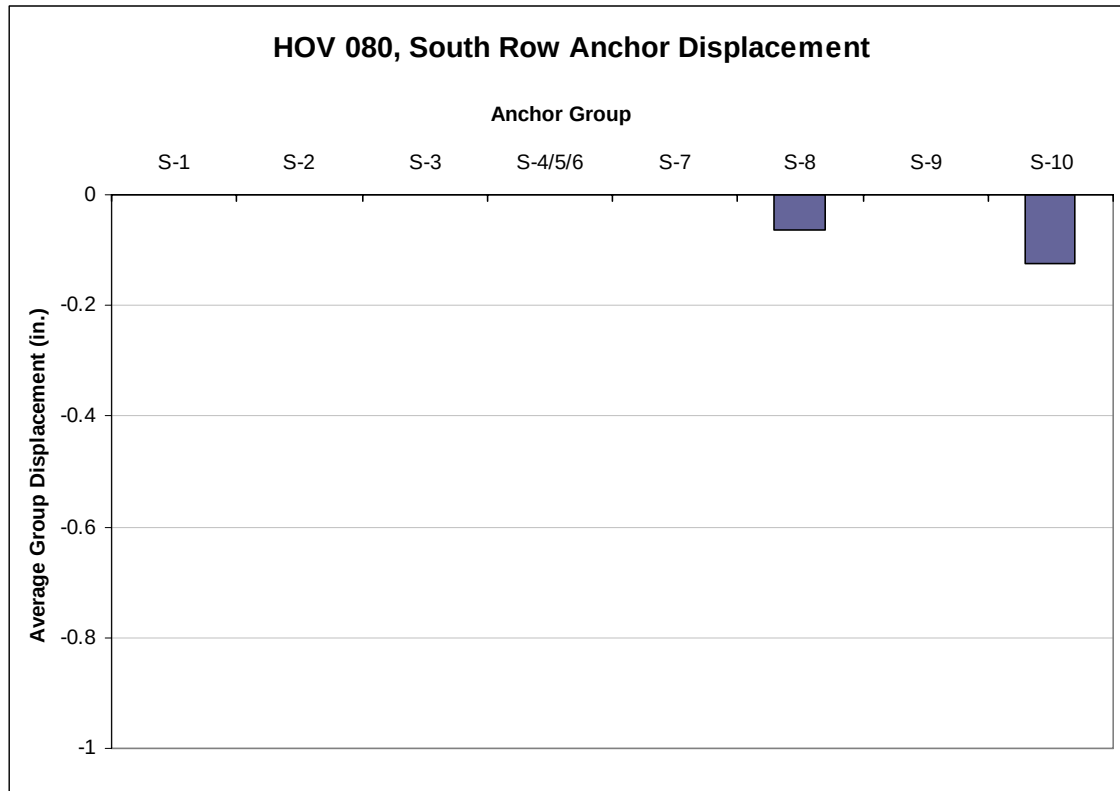
HOV 081, South Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	<i>N</i>	<i>S</i>	<i>NE</i>	<i>SE</i>	<i>NW</i>	<i>SW</i>	
S-1	0	0					0
S-2	0	0					0
S-3	0	0					0
S-4/5/6			0	0	0	0	0
S-7	0	0					0
S-8	0	0					0
S-9	0	0					0
S-10	0	0					0



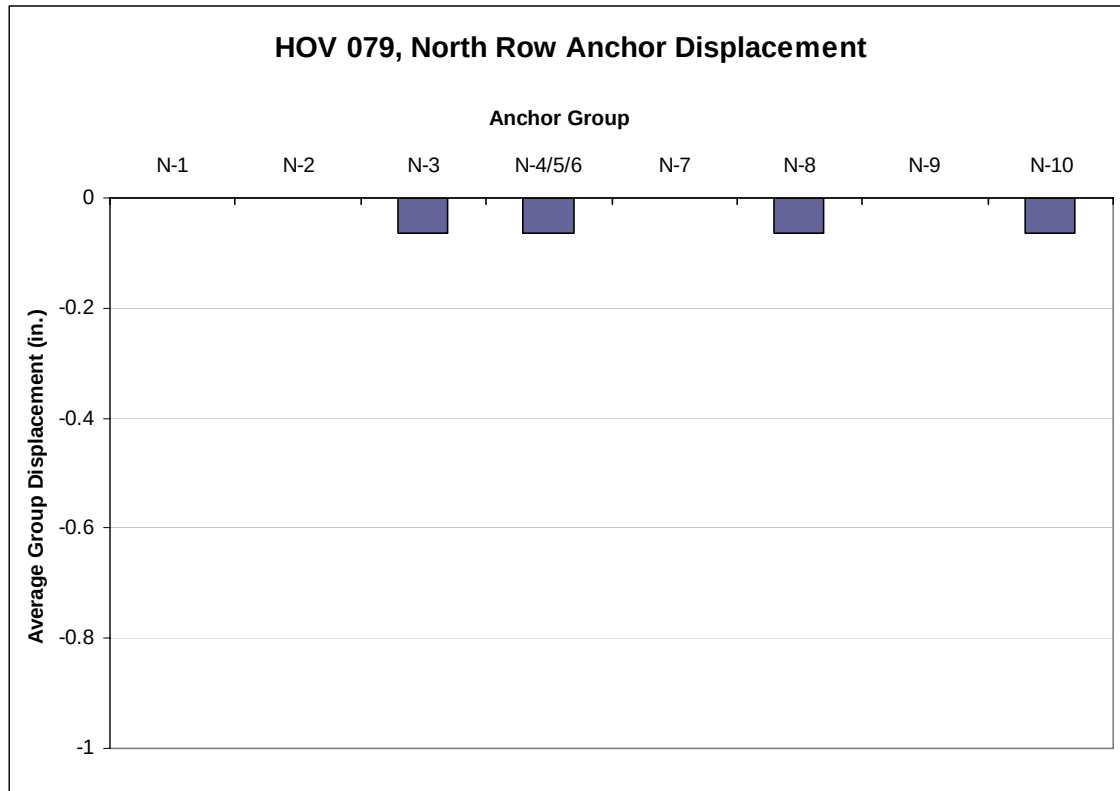
HOV 080, North Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
N-1	0	0					0
N-2	0	0					0
N-3	0	0					0
N-4/5/6			0	0	0	0	0
N-7	0	0					0
N-8	0	0					0
N-9	-0.125	0					-0.0625
N-10	0	-0.5					-0.25



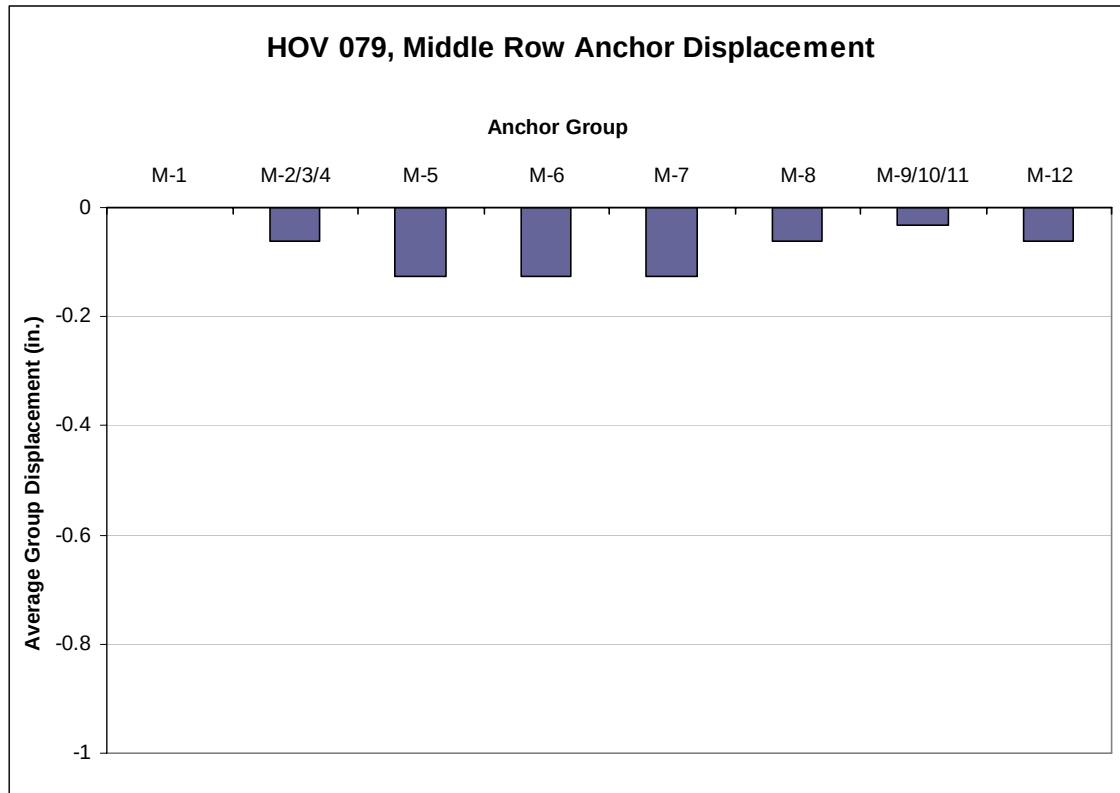
HOV 080, Middle Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
M-1	0	0					0
M-2/3/4			0	0	0	0	0
M-5	0	0					0
M-6	0	0					0
M-7	0	0					0
M-8	-0.375	0					-0.1875
M-9/10/11			0	-0.5	0	-0.5	-0.25
M-12	0	-0.625					-0.3125



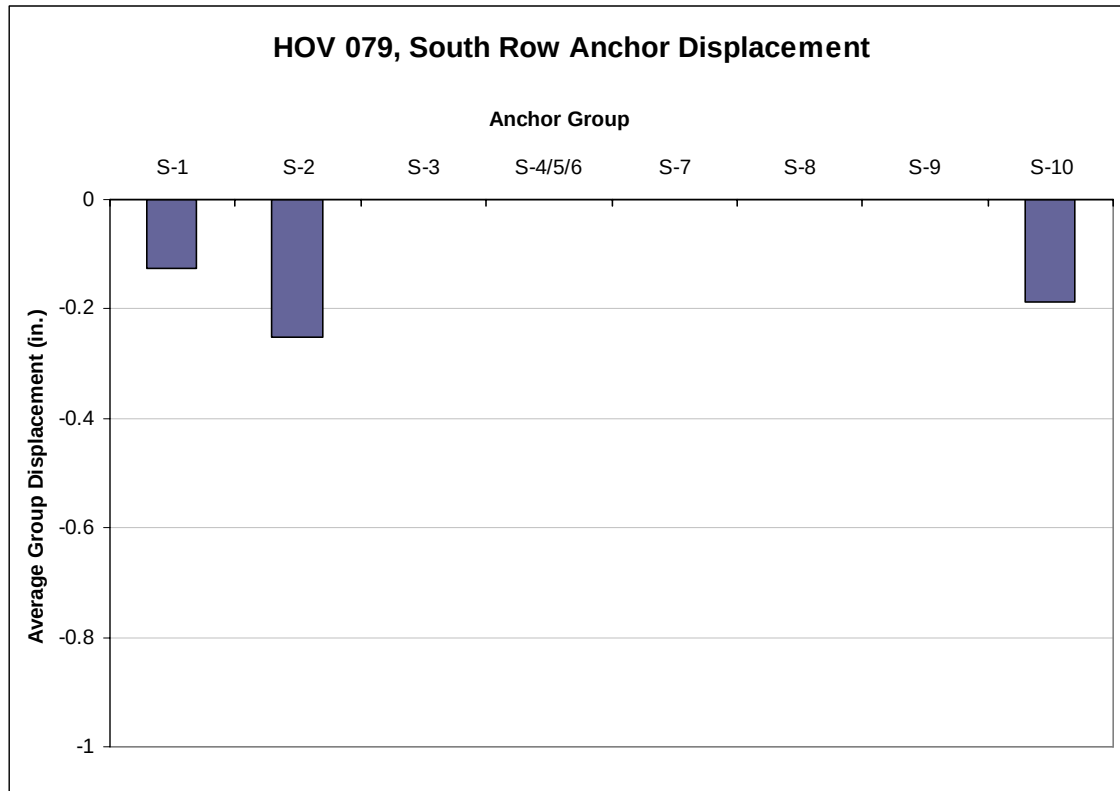
HOV 080, South Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
S-1	0	0					0
S-2	0	0					0
S-3	0	0					0
S-4/5/6			0	0	0	0	0
S-7	0	0					0
S-8	0	-0.125					-0.0625
S-9	0	0					0
S-10	-0.25	0					-0.125



HOV 079, North Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
N-1	0	0					0
N-2	0	0					0
N-3	-0.125	0					-0.0625
N-4/5/6			0	0	-0.125	-0.125	-0.0625
N-7	0	0					0
N-8	0	-0.125					-0.0625
N-9	0	0					0
N-10	0	-0.125					-0.0625



HOV 079, Middle Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
M-1	0	0					0
M-2/3/4			0	0	0	-0.25	-0.0625
M-5	0	-0.25					-0.125
M-6	-0.25	0					-0.125
M-7	-0.25	0					-0.125
M-8	-0.125	0					-0.0625
M-9/10/11			-0.125	0	0	0	-0.03125
M-12	-0.125	0					-0.0625



HOV 079, South Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	N	S	NE	SE	NW	SW	
S-1	0	-0.25					-0.125
S-2	0	-0.5					-0.25
S-3	0	0					0
S-4/5/6			0	0	0	0	0
S-7	0	0					0
S-8	0	0					0
S-9	0	0					0
S-10	0	-0.375					-0.1875

HOV 078, North Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	<i>N</i>	<i>S</i>	<i>NE</i>	<i>SE</i>	<i>NW</i>	<i>SW</i>	
No Label	0	0					0
N-1	0	0					0
N-2	0	0					0

HOV 078, Middle Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	<i>N</i>	<i>S</i>	<i>NE</i>	<i>SE</i>	<i>NW</i>	<i>SW</i>	
M-1	0	0					0
M-2/3/4			0	0	0	0	0
M-5	0	0					0

HOV 078, South Row Anchor Displacement							
Anchor Group	Anchor Displacement by Location (in.)						Average Group Displacement
	<i>N</i>	<i>S</i>	<i>NE</i>	<i>SE</i>	<i>NW</i>	<i>SW</i>	
S-1	0	0					0
S-2	0	-0.125					-0.0625
S-3	0	0					0